

Frederico Pinheiro

Software Developer

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SUMMARY

I'm a software developer with 3+ years of experience in the aerospace industry, currently focused on building flight simulators and mechanical training systems. I work mainly with C++ and Unreal Engine, developing real-time interactive environments and behaviors, where immersive design and technical depth play a crucial role. Known for a positive, calm and collected problem-solving mindset, always seeking continuous learning, and driven by a great passion for games and game development. Now, I am looking to combine my passion with my experience in interactive 3D simulations by pursuing a position in the Game Development industry.

TECHNICAL SKILLS & SOFT SKILLS

Languages: C++, C#, OpenGL, Java

Frameworks: Unreal Engine 5, VBS4, Unity, Docker, Angular

Developer Tools: GitHub, GitLab, TortoiseSVN

Modeling systems: UML, StarUML

Soft skills: Team work, Problem solving, Creative, Ambitious, Resilient, Self-motivated

EXPERIENCE

Software Developer

December 2022 - Now

Empordef Tecnologias de Informação, S.A - Website: <https://eti.pt/> .

Portugal

- Unreal Engine (C++) developer specializing in advanced training simulators for the defense sector (NATO).
- Developed an Image Generator in Unreal Engine 5 (C++), focused on an open-world environment designed for tactical training.
- Built a C++ Communication system based on the CIGI protocol to manage and control a custom Unreal Engine Image Generator in real time.
- Coordinated and collaborated with cross-functional teams to ensure seamless integration between the simulator and the Image Generator.
- Integrated external simulator flight data to execute flight routes, capture terrain imagery, and export captured data.
- Involved in projects related to aerial and ground defense technologies.
- Developed an application to control and learn the inner workings of an aircraft engine.

PROJECTS

Game Engine | C++, OpenGL, Optimization -

Jan 2026 – Now

- Designing and developing a 3D game engine from the ground up.

Through the Storm | Unity3D, C#, Project Management, Level Design

May 2020 – June 2020

- Developed the game's concept, design and defined its core technical logic and architecture.
- Implemented features across mechanics, gameplay and environment interaction.
- Built a compute shader to generate character tracks on the snow, using HLSL.
- Designed five of the game six levels.
- Assembled all the tools and features implemented by the team, maintaining a concise and cohesive representation of the game's core concept.

3D Fluid Physics Simulator | Unity, Computational Physics, C#, Optimization

May 2020 – May 2022

- Developed a physics engine for fluids in Unity, taking into account particle physics, collisions, and the fluid pressure and viscosity.
- Created multiple scenarios to test specific fluid behaviors.
- Enhanced the application's performance and architecture, including all calculations.

EDUCATION

NOVA School of Science and Technology — FCT NOVA

Portugal

Master's degree in Computer science and Engineering

2020 - 2023

NOVA School of Science and Technology — FCT NOVA

Portugal

Bachelor's degree in Computer science and Engineering

2017 - 2020

CERTIFICATIONS

Advanced Unreal Engine Developer - ArCan Studios

OTHER

Volunteer

- I volunteered at the ExpoFCT event, welcoming and assisting the visiting students of NOVA School of Science and Technology with activities in the field of interaction and augmented reality (AR).